Model Risk Factory

Evidence and human review throughout the model lifecycle

Legacy SAS / R code becomes documented Python scoring code with a tested reference.

Documentation approval and validation sign-off remain human decisions.

The operational problem

Legacy models spread evidence across code, spreadsheets and review documents

Reviewers need to connect each claim and number to its source.

Code migration needs record-level agreement with an independently supplied reference.

Monitoring findings must return to an accountable reviewer.

The model lifecycle

One workflow connects documentation, migration and monitoring

Stage	Result	Human decision
Identify and document	Inventory, cited document, rating	H1 documentation approval
Rebuild and validate	Python code, parity certificate, validation report	H2 validation sign-off
Monitor	Quality checks, drift alerts, periodic reports	Review drift and revalidation

Application architecture

A single API process coordinates agents, tools and persistent evidence

Streamlit and REST clients submit asynchronous jobs to the LangGraph workflow.

MCP tools parse code, retrieve knowledge, profile data, score candidates and compute metrics.

SQLite / Cosmos store records. Azure Files or managed-identity Blob storage hold artefacts.

Azure Container Apps runs one API replica, with a scheduled monitoring trigger.

Python rebuild and parity

Each candidate must match supplied reference scores before certification

The agent attempts LLM translation and can fall back to transcription of exported regression parameters.

A restricted scoring subprocess compares each reference record using the configured tolerance.

Mismatch feedback returns to the next attempt. The certificate identifies the code hash.

Current SAS/R fixture scores use Python reproduction. Independent legacy-runtime exports remain pending.

Validation evidence

Stored tool results supply every reported number

Dimension	Evidence
Consistency	Record-level parity against supplied reference scores
Performance	AUC, Gini, KS, Brier and calibration where applicable
Stability	Score PSI and feature distribution checks
Fairness	Segment approval rates, impact ratios and error rates

Monitoring and revalidation

A failed batch produces a visible finding and an accountable next step

Quality failures reject the batch before model scoring.

Amber and red drift generate alerts. Red drift returns the workflow to validation and a new H2 review.

Scheduled reports preserve batch metrics and their sources.

The demo's drift batches are synthetic and must be described as such.

Business case measurement

Savings need a measured manual baseline and deployment evidence

Record manual documentation and migration effort separately from human approval waiting time.

Measure agent processing time, retries and tokens on the intended real LLM configuration.

Compute portfolio scenarios from measured per-model inputs and label them as projections.

No measured manual baseline is available, so this deck makes no savings or ROI claim.

Responsible AI and review boundaries

The system assists reviewers and preserves evidence for their decisions

Input guards screen common personal-data and prompt-injection patterns, with known blind spots.

The document rater checks citations and vetoes invented variables or unsupported numbers.

H1 and H2 record a human actor. Evaluation auto-approvals are labelled simulation events.

Real LLM calibration, fairness interpretation and policy approval require owner sign-off.

Current readiness and next steps

Runtime implementation is testable. Final acceptance still needs external evidence

Available	Still required
Real agent modules and offline regression tests	Deployed real LLM calibration and end-to-end evaluation
Rebuild UI, scoring code and parity certificate	Data-owner review and independent legacy reference exports
Review deck, architecture and RAI sources	Owner sign-off, recorded video link and final email